

FLOWELL ISSY-LES- MOULINEAUX

IMPACT ASSESSMENT T0-T1

ELABORATOR EUROPEAN PROJECT

September 2025



WE OPEN THE WAY



SUMMARY

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METHODOLOGY

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ASSESSMENT T0-T1

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ASSESSMENT T0-T1

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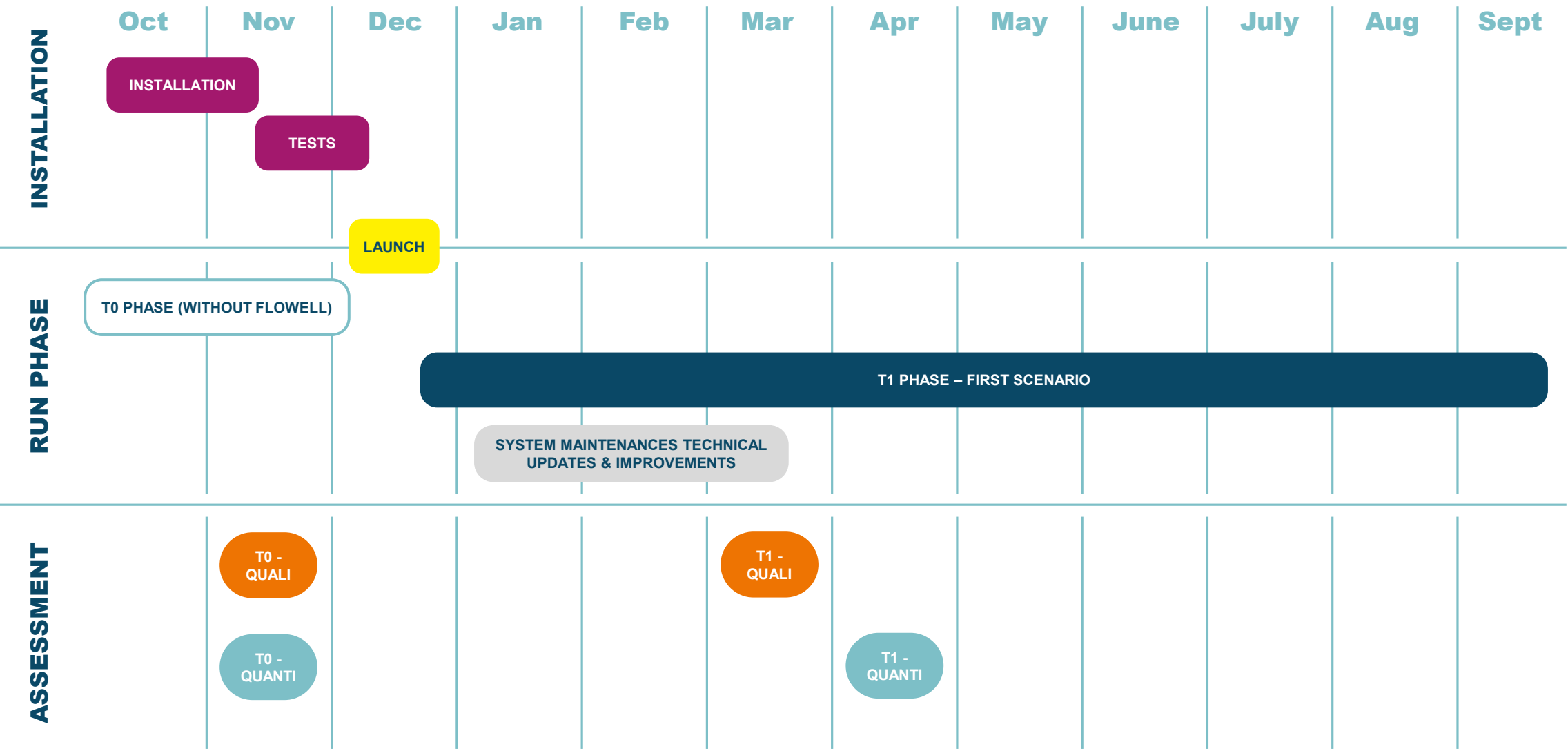




01

INTRODUCTION & METHODOLOGY

PROJECT PLANNING 2024-2025



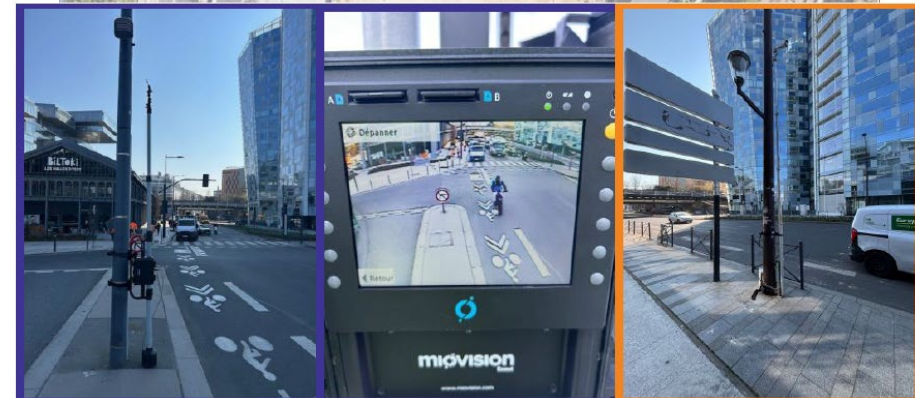


02

**QUANTITATIVE
ASSESSMENT
(COUNTS AND
ANALYSES BY
CAMERAS)**

EQUIPMENT FOR THE ASSESSMENT

- The video recording of t0 was carried out by Colas on November 26, 28, 30, and 31, 2024.
- Despite technical issues, the recording of T1 managed to retrieve 48 hours of videos corresponding to April 8, 10, 12, and 13 between 07:00 and 19:00.



*On-site temporary
cameras during the
studies*



TRAFFIC BY MODES



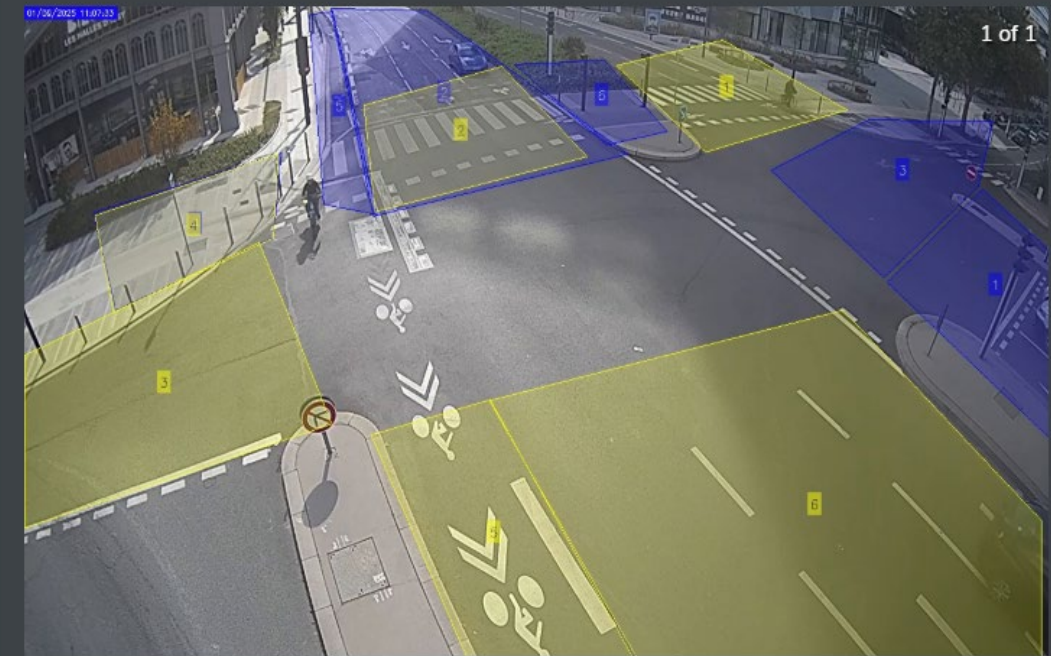
TRAFFIC

TRAFFIC ANALYSIS OF THE SITE

- The traffic was recorded during the two assessment periods : weeks T0 (week of November 25, 2024) and T1 (week of April 7, 2025).
- The counting was performed by the cameras present on the site using the automated system WINTICS.

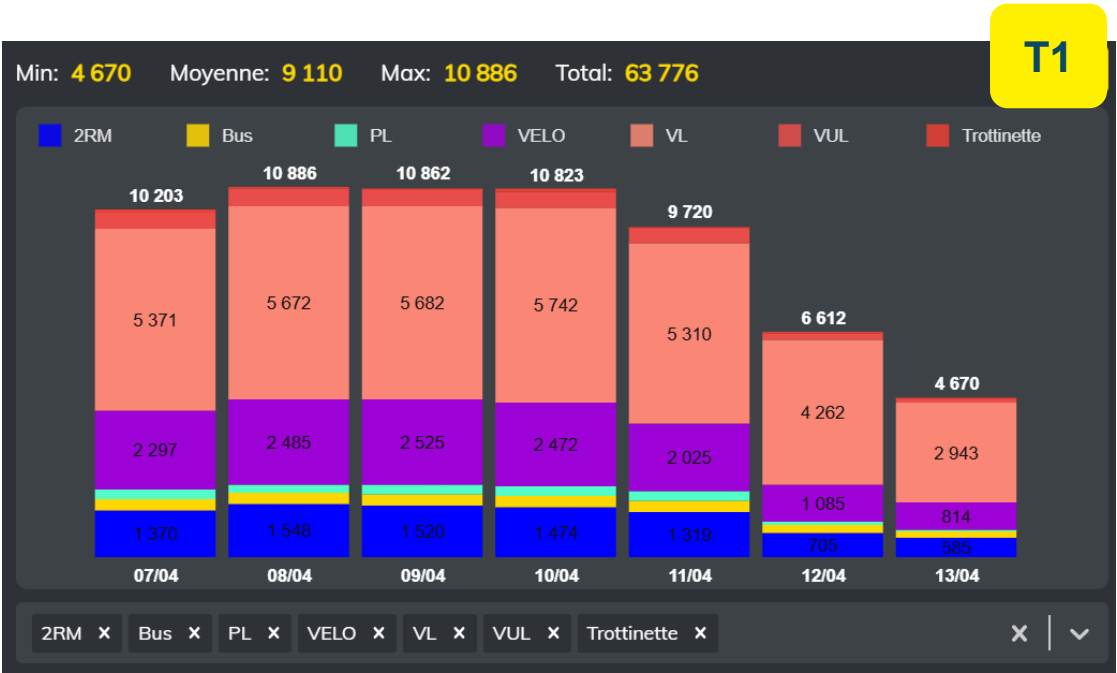
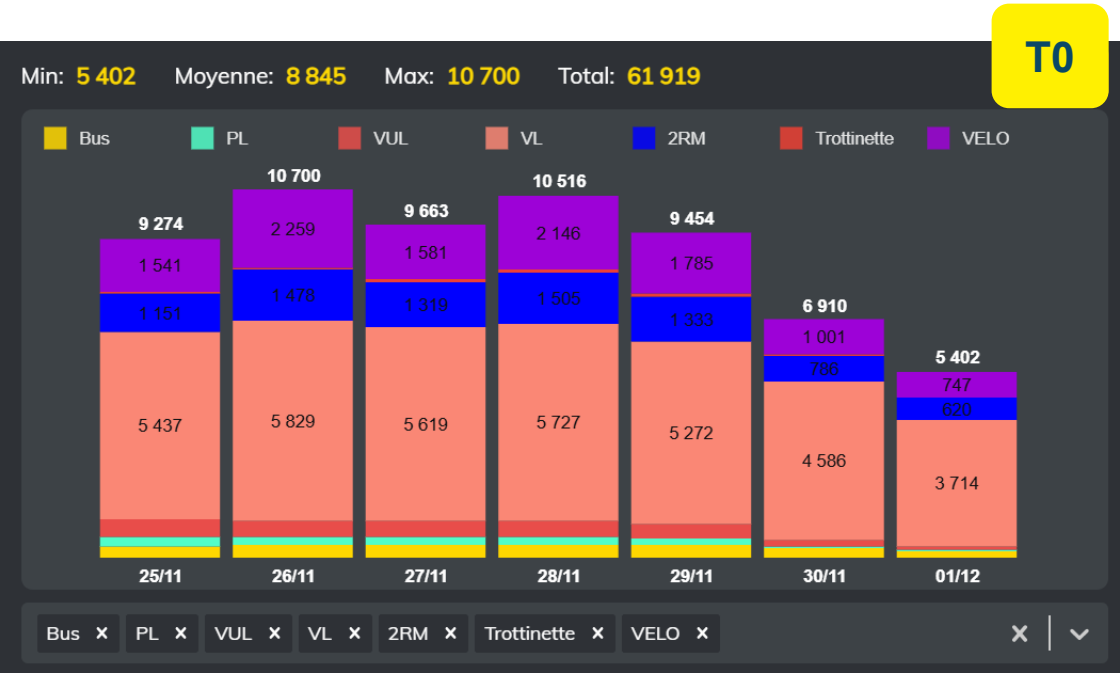
Schéma des zones

■ Zones de début de détection ■ Zones de comptage



TRAFFIC

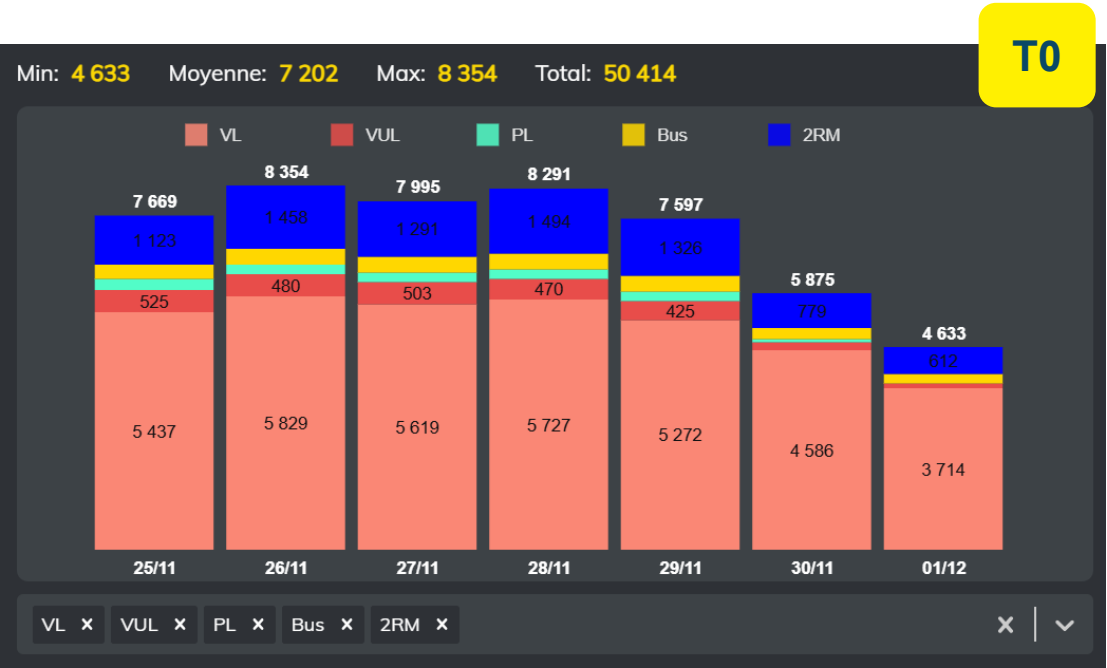
ALL TRAFFIC FROM THE DEPARTMENTAL ROAD D50



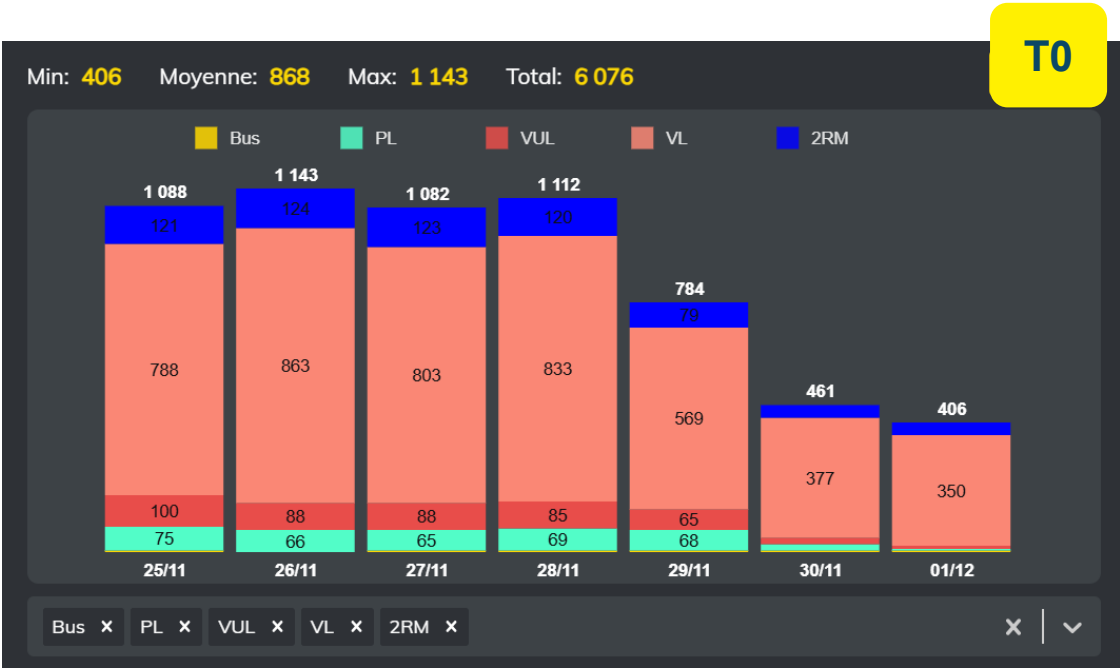
- There is a general increase in traffic of 3% with significant fluctuations in certain categories of vehicles, including a slight increase in light commercial vehicles (VUL), and a significant decrease in terms of two-wheeled vehicles (2RM). The higher average and maximum figures in T1 indicate more intense traffic compared to T0.

TRAFFIC

NUMBER OF VEHICLES GOING STRAIGHT OR TO THE RIGHT



Number of vehicles from D50 going straight or right

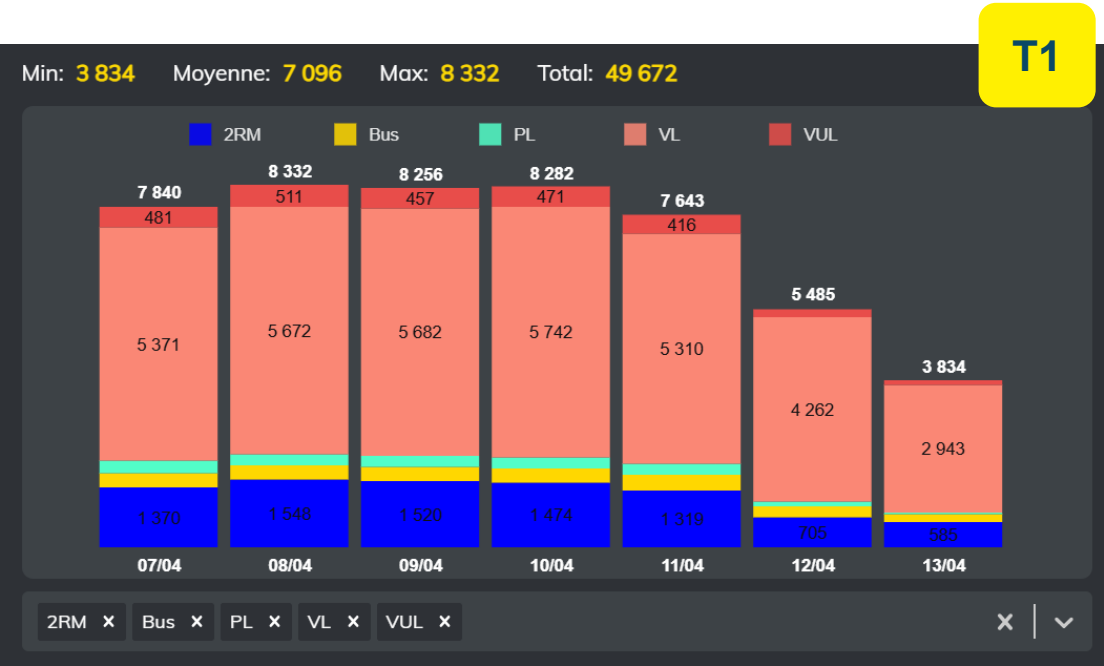


Number of vehicles from D50 turning right

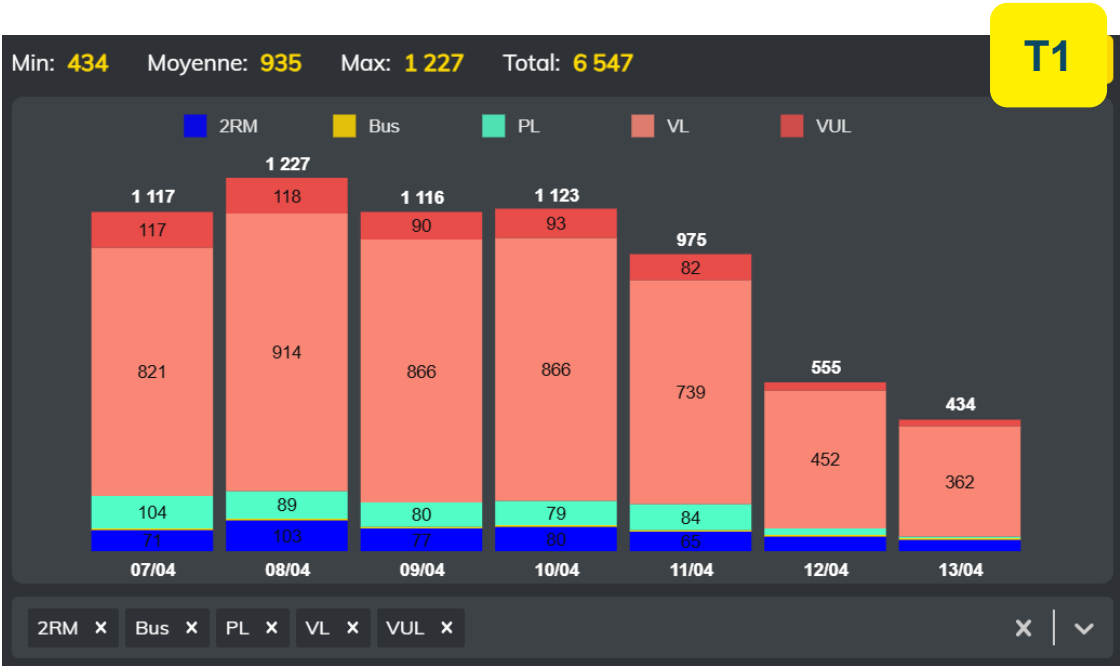
- The average ratio of vehicles turning right compared to the whole is approximately 12.05% at T0.

TRAFFIC

NUMBER OF VEHICLES GOING STRAIGHT OR TO THE RIGHT



Number of vehicles from D50 going straight or right

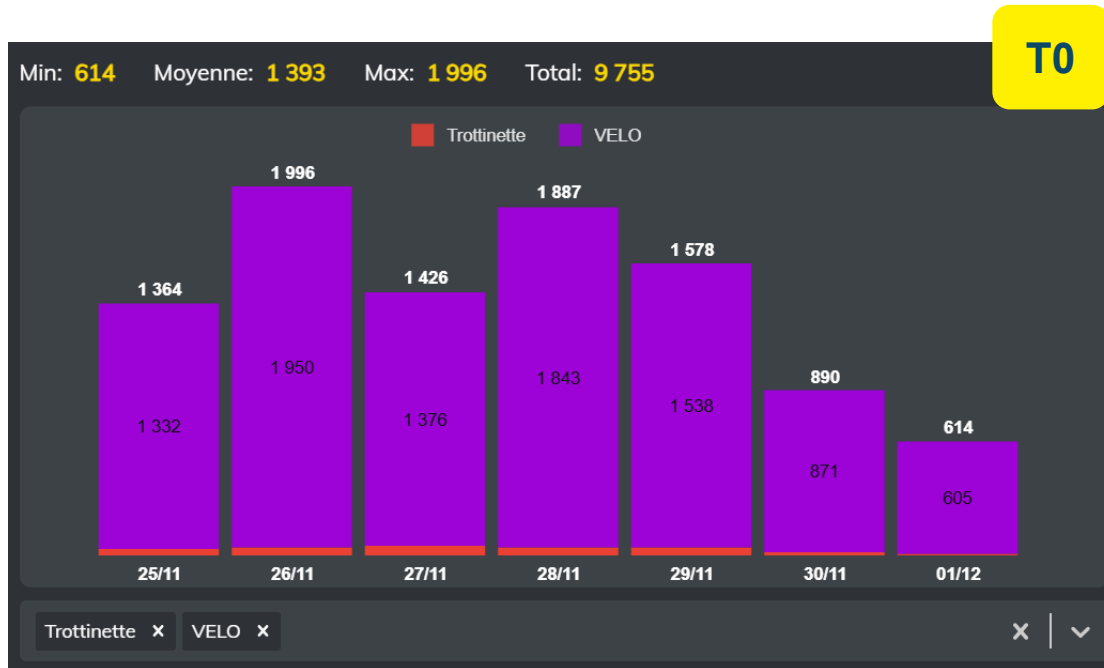


Number of vehicles from D50 turning right

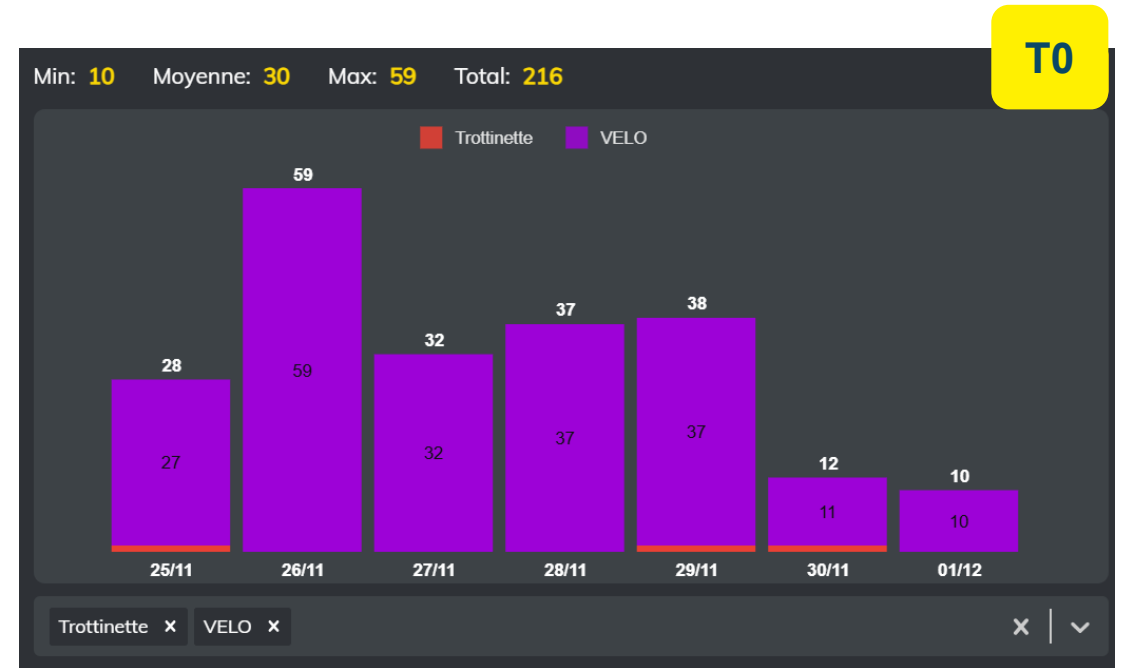
- On average, about 13.18% of vehicles turn right at T1 compared to all vehicles going straight or to the right.

TRAFFIC

NUMBER OF BICYCLES GOING STRAIGHT OR TO THE RIGHT



Number of bicycles from D50 going straight or right

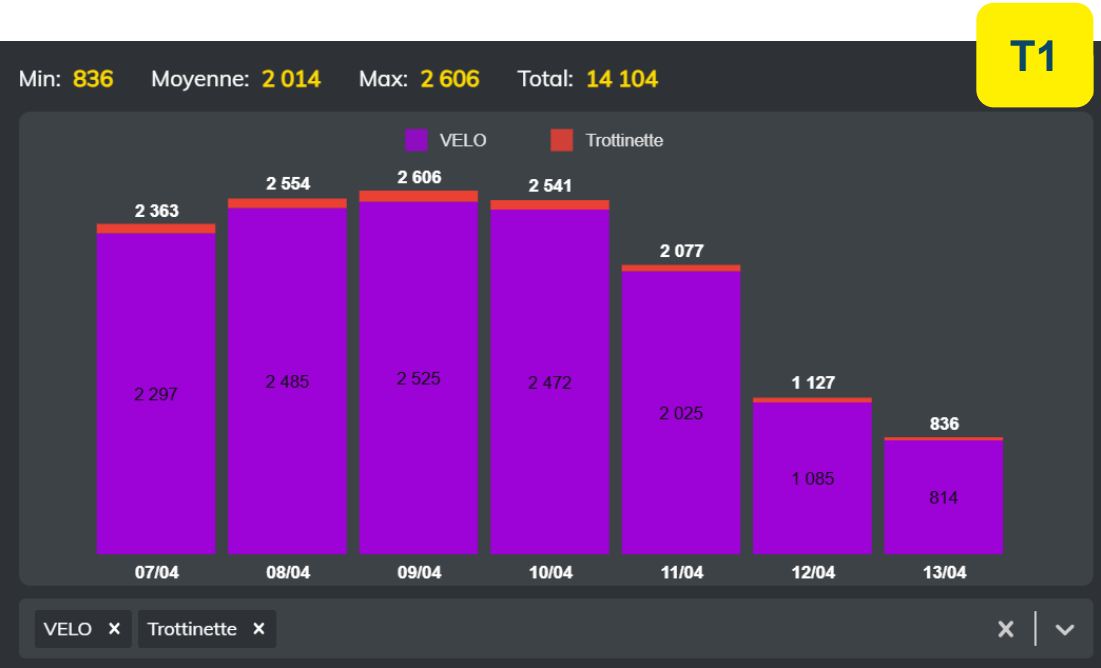


Number of bicycles from D50 turning right

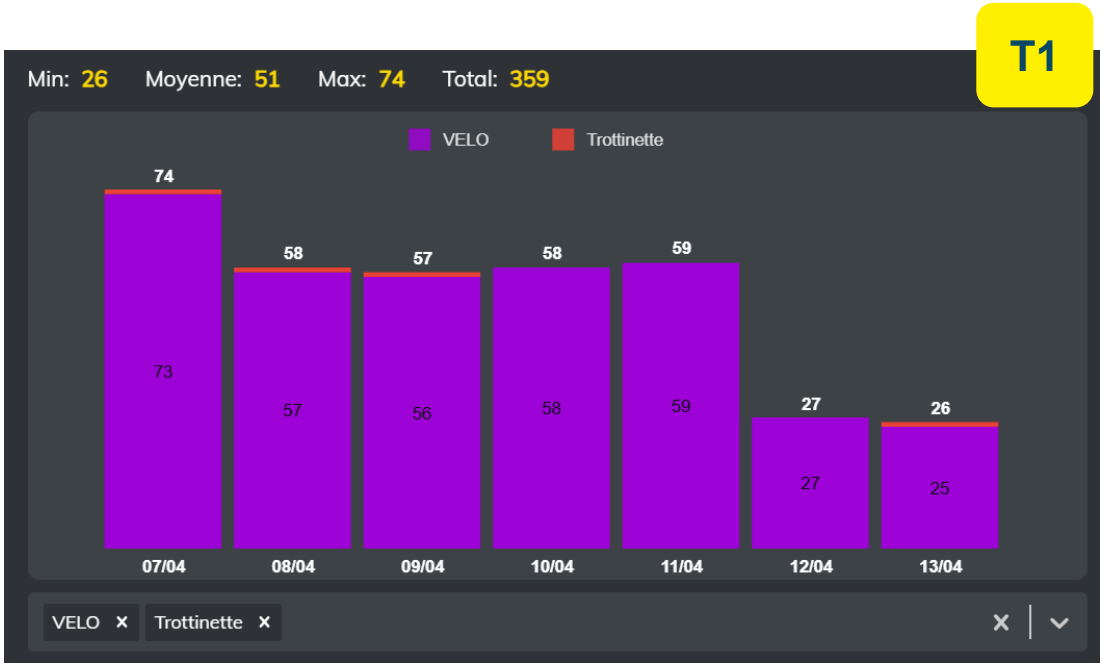
- The number of bikes turning right represents about 2.21% of the total number of bikes going straight or to the right over the given period.

TRAFFIC

NUMBER OF BICYCLES GOING STRAIGHT OR TO THE RIGHT



Number of bicycles from D50 going straight or right



Number of bicycles from D50 turning right

- On average, about 2.64% of cyclists turn right at T1 compared to all cyclists going straight or to the right.



RESULTS

Trajectories

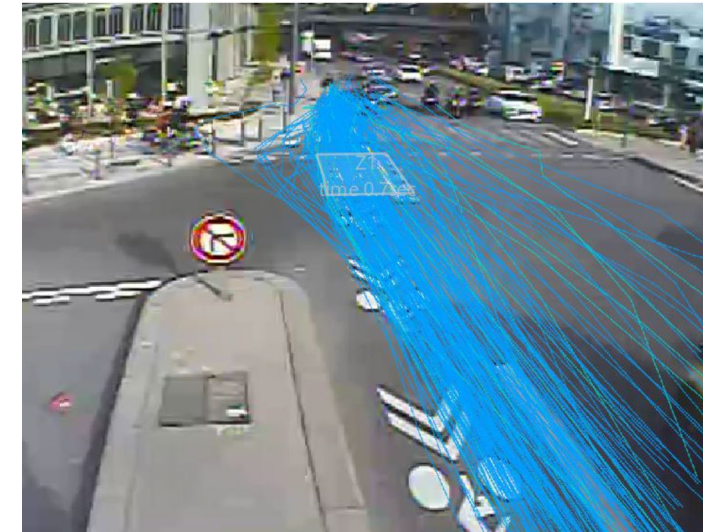
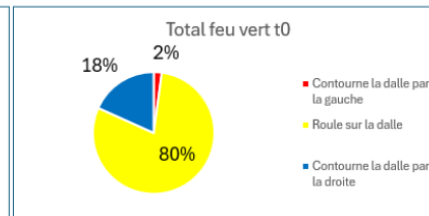
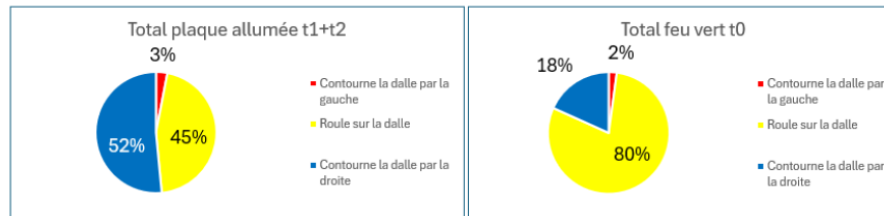
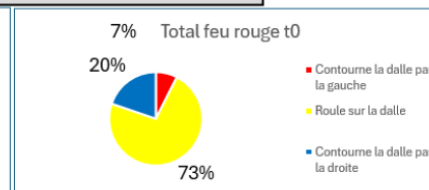
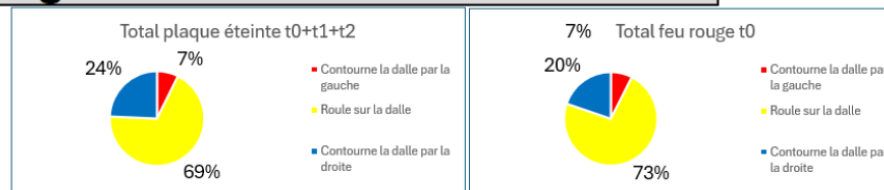


CYCLISTS TRAJECTORIES

Trajectoire des cyclistes au regard de la dalle

dalle éteinte	Contourne la dalle par la gauche	Roule sur la dalle	Contourne la dalle par la droite
t0	25	243	66
t1	23	204	93
t2	0	0	0
Total	48	447	159

dalle allumée (feu vert dans le cas de t0)	Contourne la dalle par la gauche	Roule sur la dalle	Contourne la dalle par la droite
t0	2	74	17
t1	3	45	51
t2	0	0	0
Total t1+t2	3	45	51



- When the panel is off at a red light, at T0 or T1, an average of 71% of users drive on the panel, between an average of 29% navigate around the panel to the right or to the left;
- When the panel is on at a green light at T1, more than half of users this time navigate around the panel to the right, and half drive over it;
- When the panel is not lit but the light is green at T0, we obtain approximately the same proportions as when the panel is off at a red light at t0 and t1.

Users of active modes seems to avoid the slab on the right when it activates, but this behavior does not indicate anticipation of motorists' right turns, as it doesn't occur during green light crossings.

VEHICLES TRAJECTORIES





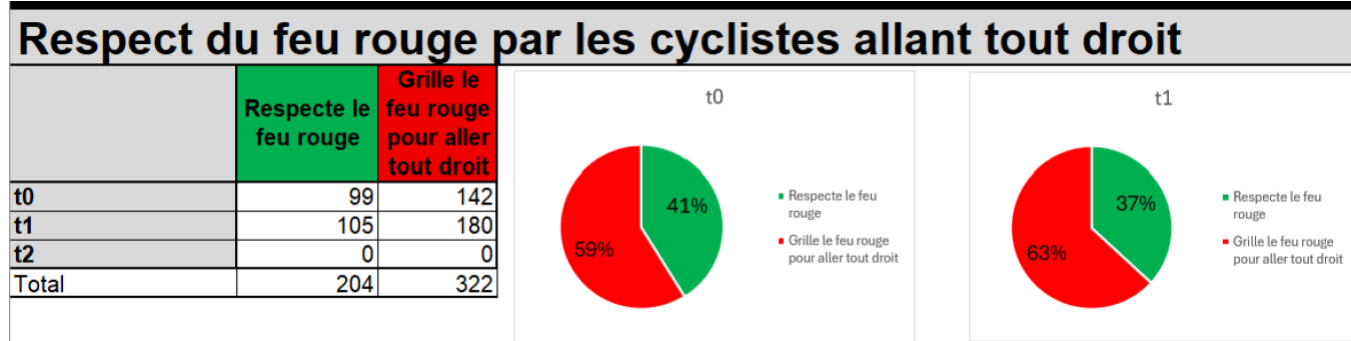
RESULTS

**Behaviors
towards traffic
lights**



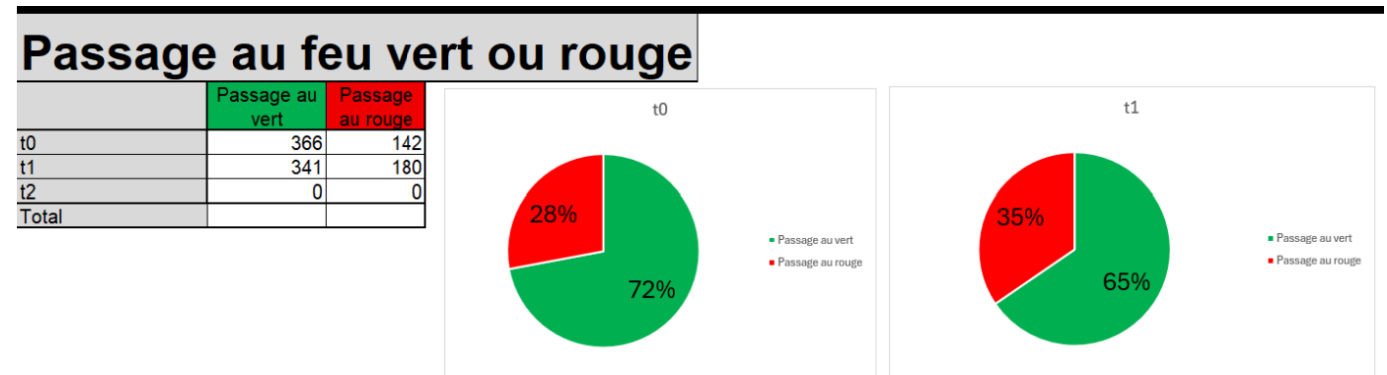
BEHAVIORS TOWARDS TRAFFIC LIGHTS

CYCLISTS ANALYSIS



- The light road marking does not seem to have any impact on cyclists' compliance with traffic lights, as **more than half of the users of the bike lane cross the intersection while facing a red light during both time periods** (59% for T0; 63% for T1).

- A small variation (not significant in our view) seems to indicate better compliance with the traffic lights at T0. This variation can be explained by a larger number of users at T1, or that the users at T0 (in November) are cyclists more familiar with this mode of transport and therefore more cautious. **Nevertheless, more than 2/3 of cyclists cross the intersection during the green light phase.**





RESULTS

**Quality of
interactions**

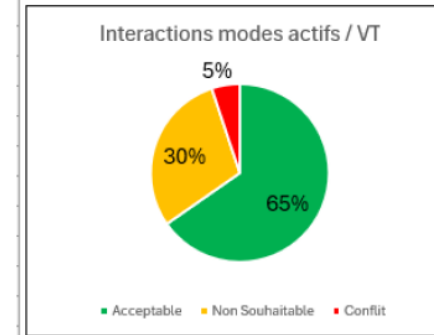
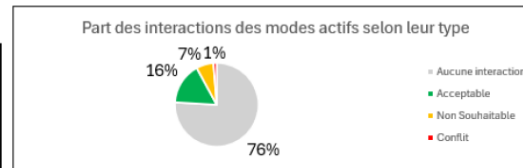


QUALITY OF INTERACTIONS BETWEEN MODES

INTERACTIONS ACCORDING TO THE TRAFFIC LIGHTS PHASE OF USERS OF ACTIVE MODES

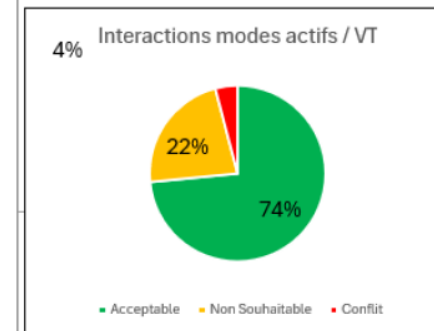
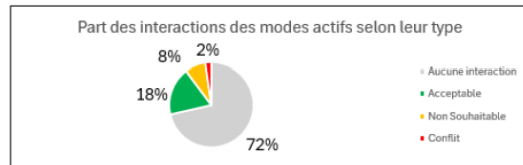
t0	Total usagers	Usagers modes actifs	Vélo	Vélo cargo	EDPM	Usagers motorisés	VT	VU	PL	BUS	2RM
Flowell	701	507	465	2	40	194	166	10	5	1	12

Interaction mode actif avec →	VT	VU	PL	BUS	2RM	Total
Aucune interaction						363
Acceptable	64	7	1	0	5	77
Non Souhaitable	29	1	1	0	1	32
Conflit	5	1	0	0	0	6
Total	98	9	2	0	6	115
Taux de rencontre	0	0	0	0	0	0
Ratio non souhaitable	29,59183673	11,11111111	50	0	16,6667	107,37
Ratio conflit	5,102040816	11,11111111	0	0	0	16,2132



t1	Total usagers	Usagers modes actifs	Vélo	Vélo cargo	EDPM	Usagers motorisés	VT	VU	PL	BUS	2RM
Flowell	742	521	451	34	36	221	158	21	14	0	28

Interaction mode actif avec →	VT	VU	PL	BUS	2RM	Total
Aucune interaction						370
Acceptable	72	5	4	0	14	95
Non Souhaitable	22	8	0	0	11	41
Conflit	4	4	0	0	4	12
Total	98	17	4	0	29	148
Taux de rencontre	0	0	0	0	0	0
Ratio non souhaitable	22,44897959	47,058824	0	0	37,931	107,439
Ratio conflit	4,081632653	23,529412	0	0	13,7931	41,4041



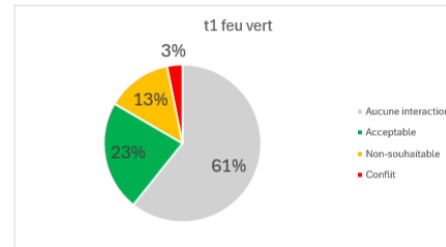
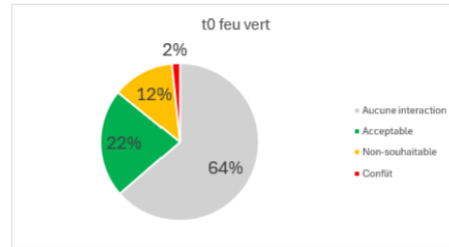
The quality of interactions observed between users of active modes and users of motorized vehicles shows little difference between t0 and t1.

QUALITY OF INTERACTIONS BETWEEN MODES

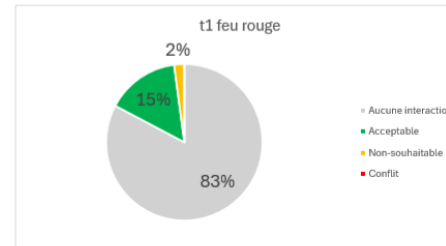
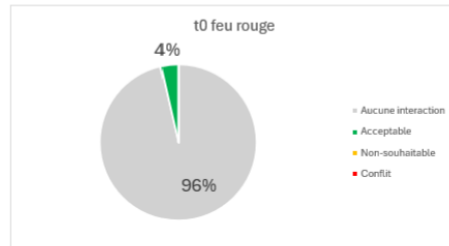
INTERACTIONS ACCORDING TO THE TRAFFIC LIGHTS PHASE OF USERS OF ACTIVE MODES

Interactions selon la phase de feu

Passage au vert	t0	t1	t2
Aucune interaction	229	212	0
Acceptable	80	79	0
Non-souhaitable	45	47	0
Conflit	6	11	0
Total	366	341	0



Passage au rouge	t0	t1	t2
Aucune interaction	134	158	0
Acceptable	5	29	0
Non-souhaitable	0	4	0
Conflit	0	0	0
Total	142	180	0



The volume varies depending on the traffic light phase at the intersection :

- Cyclists crossing on a green light experience significantly more interactions than those crossing on a red light, regardless of the type of interaction.

The users of active modes crossing on green have an interaction rate of 36% (T0) and 39% (T1), while crossing on red this ratio drops to 4% (T0) and 17% (T1).

- Interactions generally concentrate during the restart at the green light onset.
- In contrast, those who pass on a red light do so cautiously, often during the phase of 'residual red light', which allows them to cross mainly alone, similar to a 'bicycle give way' situation.

QUALITY OF INTERACTIONS

INTERACTIONS BY HOURS - OVERALL

The classifications of interactions vary according to time slots:

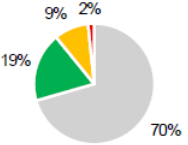
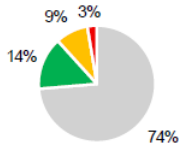
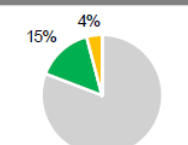
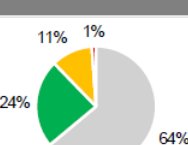
- During off-peak hours (11:00–12:00), interactions are low and considered acceptable **over 75% of the time**.
- During evening peak hours, **37% of active mode users** encounter an interaction, of which **65% are acceptable**.
- During morning peak hours, **26% experience an interaction**, but only **53% are deemed acceptable**, likely due to higher speeds and increased urgency of users.

Legend:

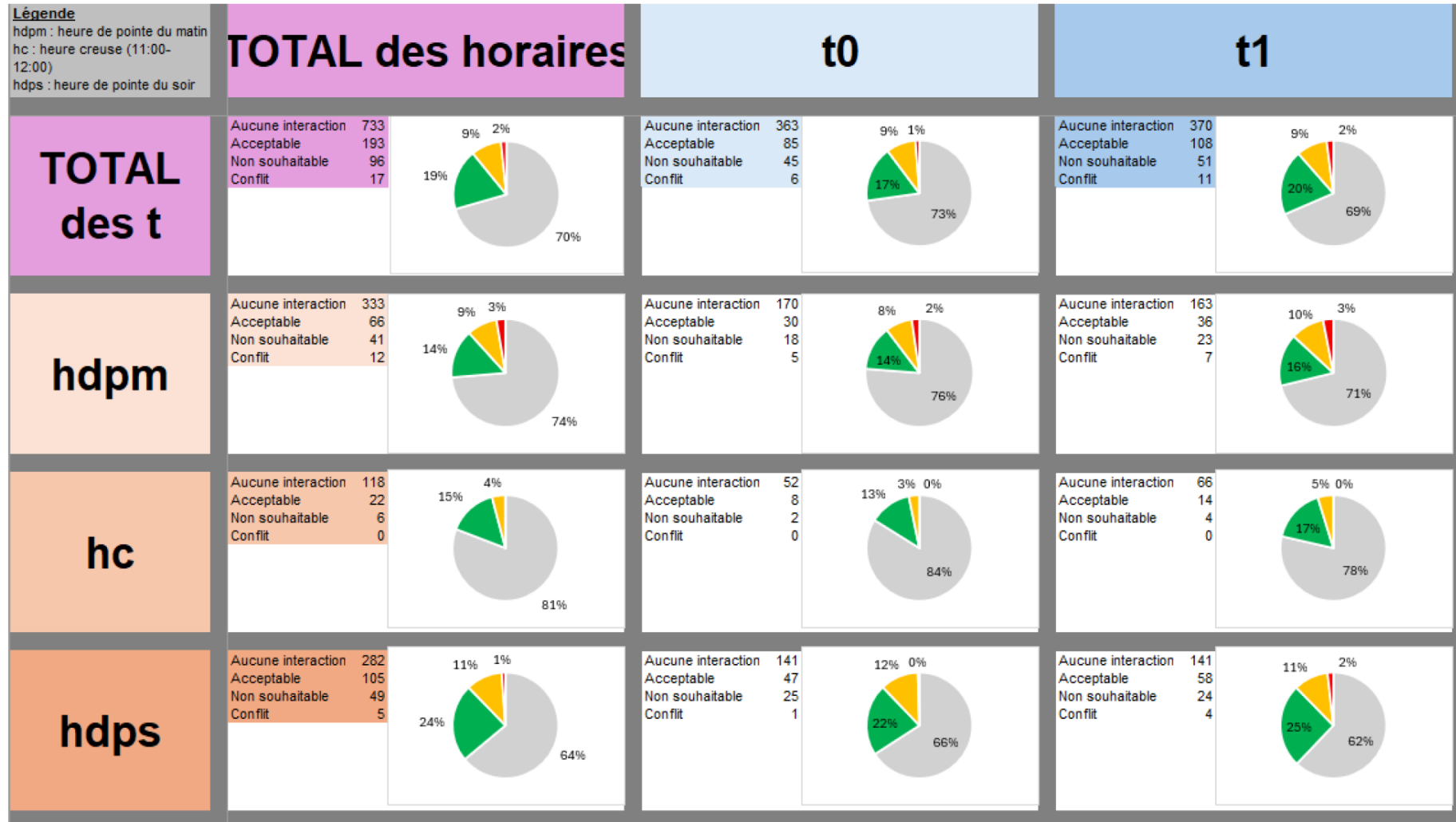
HPM : morning peak hour

HC : off-peak hour (11:00-12:00)

HPS : evening peak hour

TOTAL des t	Aucune interaction	733	
	Acceptable	193	
	Non souhaitable	96	
	Conflit	17	
hpm	Aucune interaction	333	
	Acceptable	66	
	Non souhaitable	41	
	Conflit	12	
hc	Aucune interaction	118	
	Acceptable	22	
	Non souhaitable	6	
	Conflit	0	
hps	Aucune interaction	282	
	Acceptable	105	
	Non souhaitable	49	
	Conflit	5	

QUALITY OF INTERACTIONS



Résumé des interactions des usagers des modes actifs selon la temporalité de l'étude (t0/t1) et la temporalité de la journée (matin, heure creuse ou soir)



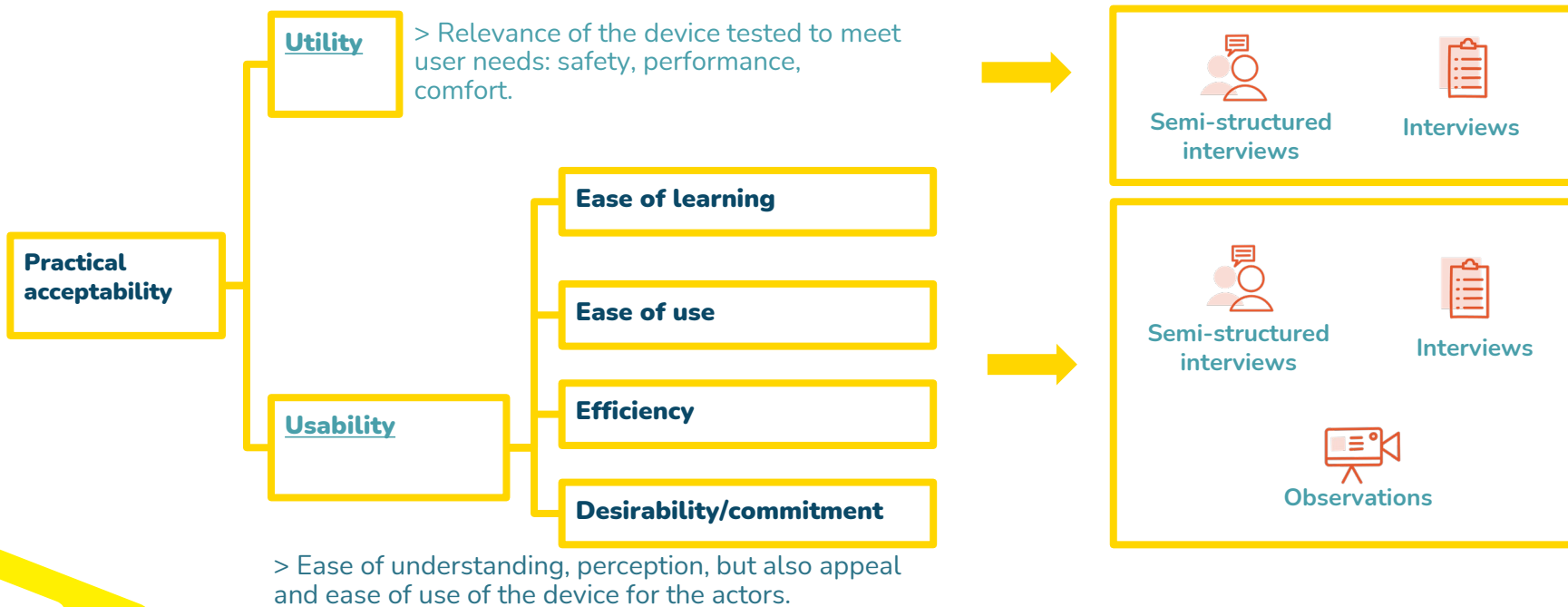
03

**QUALITATIVE
ASSESSMENT
(INTERVIEWS &
ON-SITE
OBSERVATIONS)**

QUALITATIVE STUDY

THEORICAL APPROACH

- The indicators and methods used are directly derived from the study objectives and the acceptability model. The Flowell solution will only be accepted if it is both synonymous with safety for all users (utility) and if it integrates uses.



METHOD

FROM SCREENING TO PROJECTION



T0 (DECEMBER 2024) - SCREENING

- 2 days at the Issy-les-Moulineaux site
- Observations
- Video recordings
- On-site interviews
- Videoconference interviews of approximately 30 minutes

1. Observation of user behavior at the intersection.
2. Discussions on anticipated understanding, perceived usefulness, and expected use of the system.



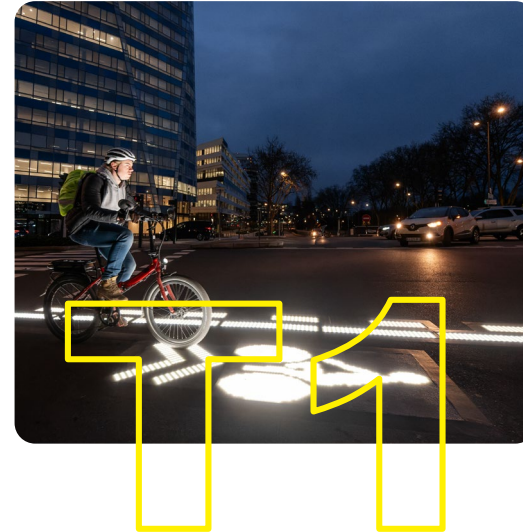
T1 (MARCH 2025) - UNDERSTANDING

- 1 day at the Issy-les-Moulineaux site
- Observations
- Video recordings
- On-site interviews
- Videoconference interviews of approximately 30 minutes

- Observation of user behavior at the intersection. Discussions focused on the effective understanding of the system, its perceived usefulness, and how it fits into daily practices.

METHOD

FROM SCREENING TO PROJECTION



T1 (MARCH 2025) - UNDERSTANDING

- 1 day at the Issy-les-Moulineaux site
- Observations
- Video recordings
- On-site interviews
- Videoconference interviews of approximately 30 minutes

Observation of user behavior at the intersection. Discussions focused on the effective understanding of the system, its perceived usefulness, and how it fits into daily practices.

OBSERVATION GRID AND INTERVIEW GUIDE

DATA COLLECTING TOOLS

- Observation grids and interview guides are essential tools for structuring and standardizing data collection, thus ensuring rigorous and reproducible analysis. They help identify key elements to observe or explore further, ensuring that all important dimensions are consistently considered.

Passage piéton						
	Qu'avez-vous compris dans cette vidéo ?	Que pensez-vous que cette signalétique lumineuse indique ?	Avez-vous trouvé l'information fournie par la signalétique claire et facile à comprendre ?	Selon vous, la signalétique attire-t-elle suffisamment l'attention sans être trop intrusive ou gênante ?	Pensez-vous que cette signalétique apporte une amélioration par rapport aux passages piétons classiques ? Pourquoi ?	Avez-vous des suggestions d'amélioration pour rendre cette signalétique plus intuitive ou efficace ?
P1						
P2						
P3						
P4						
P5						
P6						
P7						
P8						
P9						
P10						

Interview guide

[illegible]

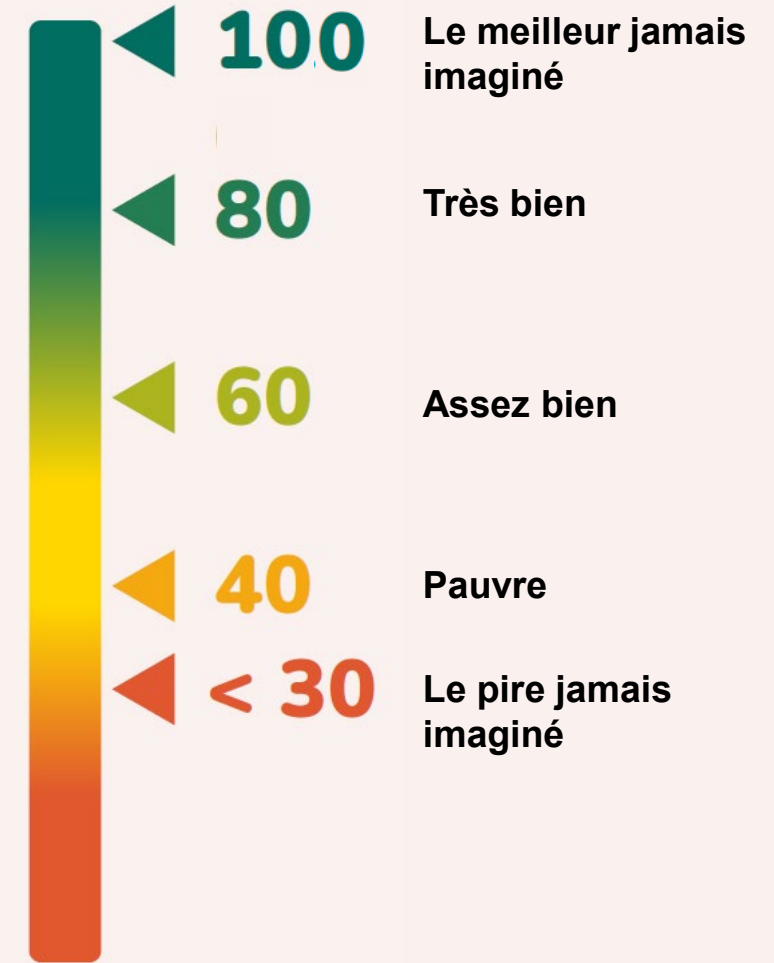
Observation grid

QUESTIONNAIRE SUS

The SUS questionnaire assesses the usefulness, usability level and ease of use of the system according to users.

	Pas du tout d'accord						Tout à fait d'accord
Je pense que je vais utiliser le service fréquemment	☐	☐	☐	☐	☐	☐	☐
Je pense que le service est inutilement complexe	☐	☐	☐	☐	☐	☐	☐
Je pense que le service est facile d'utilisation	☐	☐	☐	☐	☐	☐	☐
Je pense que je vais devoir faire appel au support technique pour pouvoir utiliser ce système	☐	☐	☐	☐	☐	☐	☐
Je trouve que les fonctionnalités du service sont bien intégrées	☐	☐	☐	☐	☐	☐	☐

- *Statements are proposed to the user to which he responds according to his level of agreement*

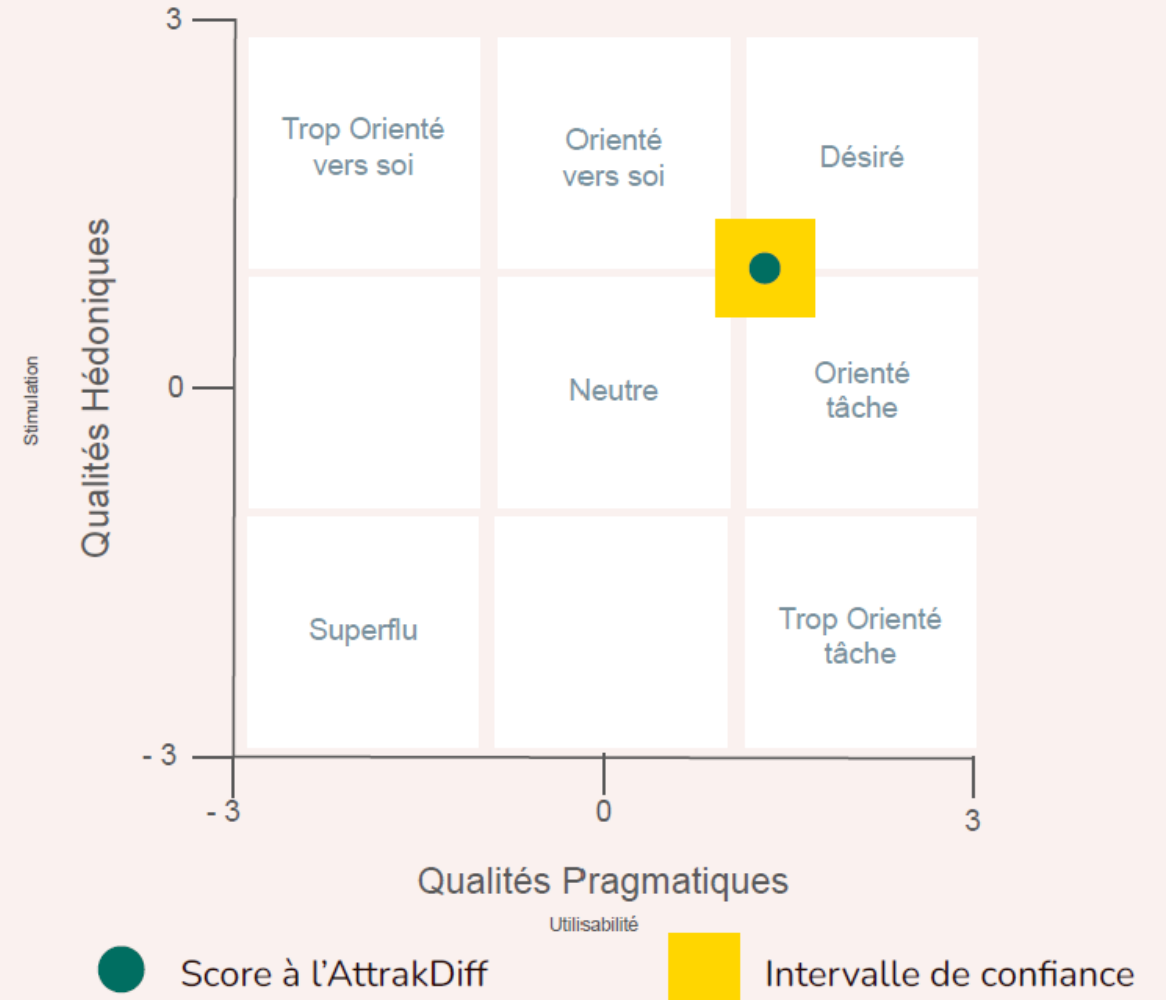


QUESTIONNAIRE ATTRAKDIFF

The AttrakDiff questionnaire used here allows for the evaluation of both the usability of the device — that is, whether it is perceived as simple, clear, and easy to use — and its experiential value, through the pleasure, interest, or stimulation it may evoke.

Technique										Humain
Compliqué										Simple
Pas pratique										Pratique
Fastidieux										Efficace
Imprévisible										Prévisible

- Pairs of contrasting words are presented to the respondents



PANEL DESCRIPTION

QUANTITATIVE ASSESSMENT T0-T1

Total
29



15 People interviewed via videoconference - T0
Testing slots of approximately 30 minutes

14 People interviewed by videoconference - T1
Testing slots of approximately 30 minutes

+ Around ten people were interviewed on-site at T0 and T1.



RESULTS

**Observed
behaviors**



BEHAVIOURS

OBSERVED BEHAVIOURS

- During peak and off-peak hours, the proportion of cyclists respecting the traffic signal has doubled between T0 and T1.
- At the same time, **the number of risky interactions between cyclists and motorists has been halved.**

Several factors can contribute to this evolution :

- **The marking of the bike lane**, redone during the implementation of the system, makes the signage more visible and understandable, which can encourage cyclists to stop more.
- During observations at peak hours at T1, **the presence of parked utility vehicles on the bike lane** forced cyclists to ride on the general roadway, placing them behind stopped vehicles and preventing them from crossing the traffic light.

Users obeying the red light

Off-peak hours
(10 a.m. – 11 a.m.)



11%
8 cyclists out of 71

25%
22 cyclists out of 88

Peak hours
(7:45 a.m. – 8:45 a.m.)



22%
58 cyclists out of 263

43%
91 cyclists out of 210



Note: 7 risky situations (between vehicles and cyclists) were observed at T0 and 6 situations at T1.

BEHAVIOURS

OBSERVED BEHAVIOURS

➤ Frequent crossing of red lights by cyclists

Slight slope making it difficult to restart, Configuration of the double light (the first turns red while the second remains green), Established habit

➤ Conflicts related to the pedestrian light (Quai du Président Roosevelt)

Frequently leads vehicles to stop on the bike lane

➤ Numerous risky interactions with vehicles coming from the opposite direction and turning left

More refusals of priority from these users Lower attention from soft mobility users

➤ Low vigilance of cyclists

Cyclists who enter on a red light are more aware of the dangerous situation and perform visual checks.





RESULTS

**Perception of
the light road
marking**



PERCEPTION

VISIBILITY OF THE LIGHT ROAD MARKING



68,3 %

of surveyed people believe that the system improves the visibility of the bike lane.

- The lighting system seems to provide a real added value compared to the current arrangements from the users' point of view.

"I would say that it's interesting because it grabs the driver's attention, it's a novelty, it draws attention."

PERCEPTION

VISIBILITY OF THE LIGHT ROAD MARKING (T1)

86 %

of the respondents agree that the device is very visible at night and attracts quite a bit of attention. However, during the day, it lacks a bit of visibility.

- These feedbacks echo those observed in previous studies conducted in other sites (Saint-Mandé, Carquefou...), despite different implantation contexts.

"I think it can be a good solution, at least at night it is quite visible, during the day I'm not very convinced."



PERCEPTION

VISIBILITY OF THE LIGHT ROAD MARKING (T1)

- Despite good overall visibility, we observe that users, both cyclists and motorists, may have difficulty seeing the device **in case of heavy traffic.**

Indeed, we have observed that in the case of dense traffic flow, cyclists focus their attention on other nearby road users (notably the different vehicles), which may lead them to neglect or partially ignore the traffic signals.





RESULTS

**Perception of
the light road
marking**



UNDERSTANDING

UNDERSTANDING OF THE LIGHT ROAD MARKING A PRIORI (T0)

- The majority of participants understood the issue that this light device seeks to address as well as its benefits in terms of safety.

- Some thought it was an elevated bike path.
Note: At T0, the work was not completed and an over-thickness was visible.

"It highlights a persistent danger: cars turning while driving over the bike lane, which are supposed to yield but do not. As a result, they can hit a bike that is in their blind spot."

*"Some thought it was an elevated bike path.
Note: At T0, the work was not completed and an over-thickness was visible. "*

- Before understanding how it works, participants generally thought they were facing a simple ground marking. **This confirms that the turned-off device is well integrated and does not attract more attention than other signage.**

UNDERSTANDING

UNDERSTANDING OF THE LIGHT ROAD MARKING A POSTERIORI (T1)

- The ignition system of the device was well understood by the majority of participants: **they understand the priority rules that the device aims to reinforce.**
- Overall, the participants stated that they would have understood the operation of the signage if they had encountered it (both as a cyclist and as a motorist).

"The ground marking is very clear, it's to facilitate the bike crossing."

"When the light turns green, it lights up the strip, so it's not bad because it seems to illuminate well, adding an extra element for the cars."



UNDERSTANDING

UNDERSTANDING OF THE LIGHT ROAD MARKING A POSTERIORI (T1)



"The ground marking is very clear, it's to facilitate the bike crossing."

78,5 %

of participants believe that this system is understandable by all users, even if it may require some adjustment time.

- Like many innovative systems, **an adaptation phase** seems necessary. Understanding and integrating the meaning of the signage may require several confrontations for users.

One way to measure this would be to compare the behaviors of users who occasionally cross this intersection and those who do so regularly.

UNDERSTANDING

UNDERSTANDING OF THE LIGHT ROAD MARKING A POSTERIORI (T1)

- On the other hand, the device does not seem to be well perceived and/or understood by drivers coming from the opposite lane and turning left.

Several risky interactions have been observed between these vehicles and cyclists on the bike lane, indicating a lack of understanding of priority or a low anticipation of their presence.



It should be noted that these vehicles do not pass directly over the Flowell device during their maneuver, as it is located outside of their trajectory.



RESULTS

**Perceived
utility of the
lighting mood
marking and
projection**



USEFULNESS

USABILITY QUESTIONNAIRE

80/100

Bicycle lane

80* ▶

*Questionnaire presented
only to T1 participants

- A score of 80 reflects a very good satisfaction score from participants and indicates a high level of usability.
- This score also confirms the feedback expressed during the interviews, in which participants repeatedly highlighted the clarity of the system and its operation.
- Participants find this system useful particularly because they believe that motorists will pay more attention to the potential presence of cyclists.

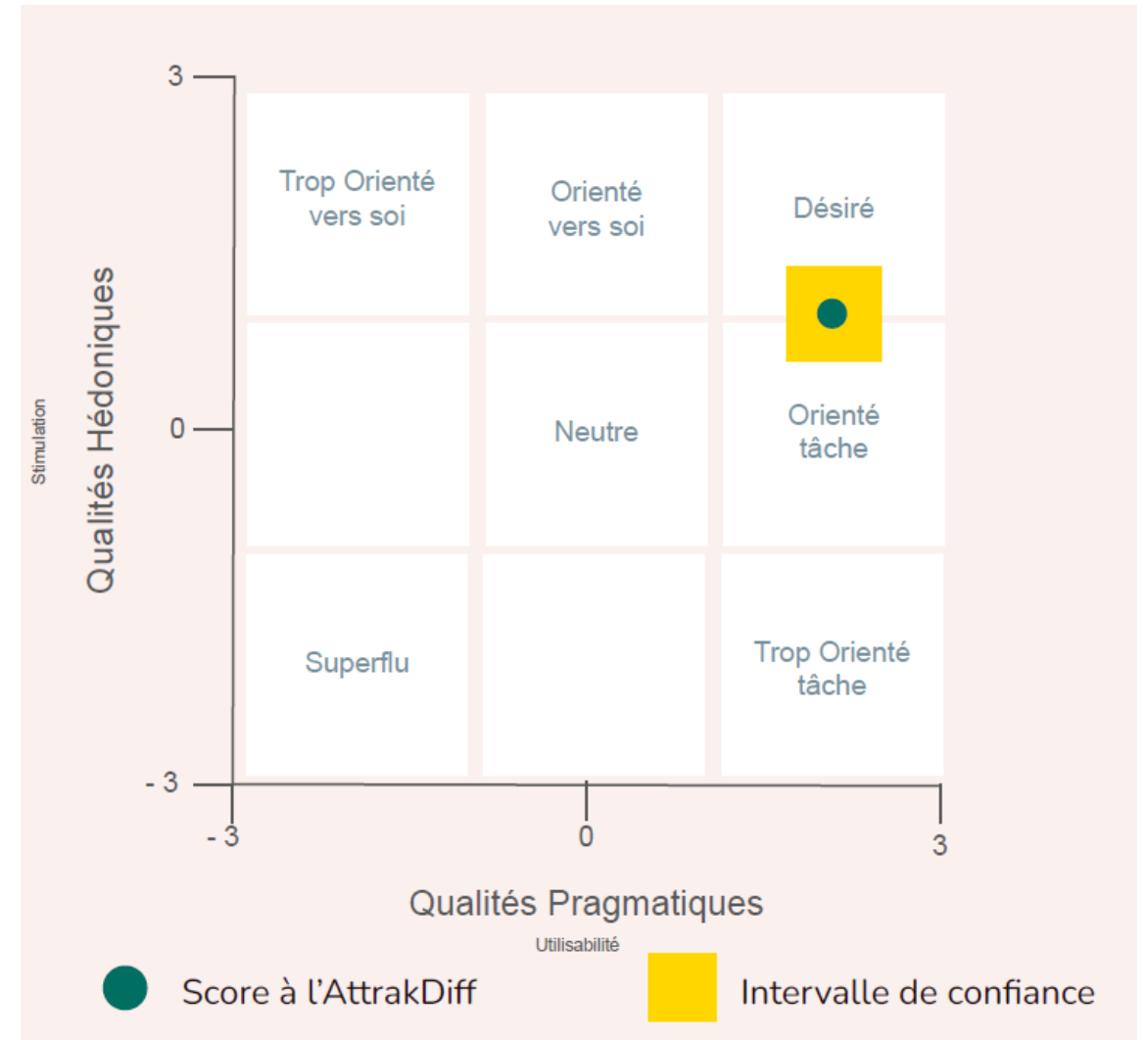
"Yes, compared to a painted yield sign, it's better."

USEFULNESS

ATTRAKDIFF QUESTIONNAIRE

- The Flowell device is perceived as useful, clear, and functional by the participants, showing a rather good perceived usability.
- Regarding user stimulation, the device is seen as pleasant, innovative, and engaging, while not overshadowing the other signage in the area in terms of visibility.
- This is a positive point in the sense that this device should not add visual pollution for users.

"We tell ourselves that we are more likely to be seen than to have our lives recognized, so we feel safer."



PERCEIVED USEFULNESS

WITH THE LIGHT ROAD MARKING (T1)

78 %

of participants believe that this device improves the safety of users of soft mobility, as they are more likely to be seen.

- Even though this device seems to improve the feeling of safety for cyclists, they admit to maintaining a level of vigilance :

"The problem is not the road markings, it's the people. I would feel more reassured but I remain on my guard."



- In general, the results of the interviews and questionnaires demonstrate a good acceptability of the system.

A larger sample would be necessary to statistically support these results. A new study, once the device is actually integrated into daily practices, could be interesting.



04

➤ **OVERALL
CONCLUSIONS T0-T1**

QUANTITATIVE SYNTHESIS

OVERALL T0 – T1 ASSESSMENT

- At this stage, it is difficult to draw clear conclusions about the role of the Flowell device in respecting the priority of active mode users by motor vehicle users, given the marginal difference observed between t0 and t1.
- It will therefore be interesting to observe behaviors and interactions at t2, during darker periods, particularly during the morning and evening peak hours which are especially busy at this intersection.
- For now, the only significant data drawn from the analysis of t0 and t1 allows us to state that:
 - active mode users voluntarily take detour routes to the right when the light is on
 - active mode users who cross the intersection on a red light, especially at the end of the red light phase, are those who have the least interactions and are the safest when crossing the intersection.



QUALITATIVE SYNTHESIS

OVERALL T0 – T1 ASSESSMENT

		T0	T1
PERCEPTION	➤ The Flowell device seems to be well perceived by motorists and users of soft mobility (highlighted through observed behaviors and projection videos). ➤ The device is deemed very visible and effective at night, but its visibility during the day is perceived as weaker, which limits its impact.		
USEFULNESS	➤ The perceived usefulness of the system is quite good both in projection at T0 and after having seen the system at T1. ➤ The system enhances the information sent to drivers regarding the potential presence of cyclists in this area.		
UNDERSTANDING	➤ The issue that this system aims to address, as well as its functioning, has been understood by the interviewed users. The understanding of the system is built gradually, and a period of adaptation seems necessary – a classic dynamic in the processes of acceptability of new technologies.		
REAL BEHAVIOURS	➤ The system being relatively new, we observe certain ingrained habits from users, particularly regarding the disregard for traffic lights. ➤ During peak hours, the environment quickly becomes congested (with vehicles parked on the bike lane), which forces cyclists to navigate around the system and thus have less contact with it.		



05

➤ **NEXT STEPS & T2
SCENARIO OPTIONS**

PATTERNS SUGGESTED FOR T2

T1

SI Feu Vert ou Orange véhicule/vélo



SI Feu rouge véhicule/vélo

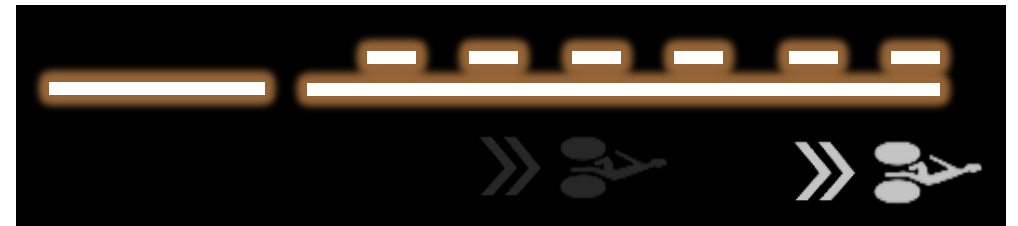


T2

SI Feu vert ou Orange véhicule/vélo & Présence de vélo



SI Feu vert ou Orange véhicule/vélo & Absence de vélo



SI Feu rouge véhicule/vélo



